Solar PV Power Plant: Favorable or not?

Diana Awimrin

11/5/19

ChE 451

Executive Summary:

The purpose of this case study is to determine the feasibility of installing a solar PV power plant to generate electricity throughout a refugee camp in the North of Jordan compared to a traditional power generation system, such as diesel electric generators. Both technical and economic feasibilities will be studied. The refugee camp considered houses 20,000 people in prefab units as shelter. These units can house anywhere from 4-6 people. The proceeding calculations and discussion will be based off housing the minimum number of people per prefab unit to account for the greatest number of prefab units the camp would have. With that said, the values calculated will be capped off at the maximum amount of energy usage and costs of the system's several components. This report will consist of a design plan, sizing analysis, and conclude with selecting either the PV solar system or diesel electric generators as the more economically favorable option. This report concluded that a PV solar power plant has a cost per kWh of \$0.14/kWh and the diesel generator has \$0.34/kWh deeming the solar PV system to be more economically favorable. Also, solar panels have far more advantages to generators, to safely conclude that it is the best option.

Sizing Analysis and Design Plan:

The solar PV power plant needs to accommodate for 20,000 people's electricity and therefore some calculations are performed. The tables below are calculated values of the variables that need to be known in order to ultimately determine the cost of the project upfront and the cost of the lifecycle, as well as the cost of the kWh for the system. Some assumptions are made and will be discussed in the sample calculations. Also, the solar panels will be placed on an area of land near the camp at a 30-degree tilt for optimum efficiency [1].

Table 1: Preliminary Data for PV Systems

Load Type	**# Units**	**Rated Power (W/unit)**	**Ppeak (W)**	**Daily Usage (hr)**	**Eused (Wh)**
Prefab Unit					
TV	1	50	50	6	300
Fan	2	40	80	14	1120
LED Lights	5	10	50	8	400
Small Fridge	1	75	75	24	1800
Clinic					
Fridge	2	200	400	24	9600
LED Lights	10	10	100	24	2400
Medical Equipment	2	1500	3000	24	72000
Others	1	500	500	24	12000
Pumping water Supply					
Electrical Motors	1	3000	3000	24	72000
Controls Panels	2	500	1000	24	24000
School Lighting					
LED Lights	50	10	500	12	6000
Others	1	500	500	24	12000
Street Lighting					
LED Lights	100	25	2500	13	32500

Table 2: Calculated Data for PV Systems

Ppeak Total (kW)	1286.5
Eused Total (kWh)	18342.5
Tsun (hr)	10
Cost of Invertor ($)	321625
Cost of Panels ($)	641987.5
Cost of Batteries ($)	9171250
Cost of batteries, Life Cycle ($)	22928125
Installation ($)	643250
Cost Upfront ($)	10778112.5
Cost Lifecycle ($)	23891737.5
Cost kwh ($)	0.142743497

Sample Calculations:

The data that needs to be calculated is the peak power usage (Ppeak), the total daily energy usage (Eused), the cost of the inverter, cost of the panels, cost of the batteries, cost of the batteries life-cycle, the cost upfront, the cost of the lifecycle and the cost of the kilowatt-hour. Calculations below are repeated for all powered areas such as the clinic, school lighting, etc. and then summed to determine the values seen on *Table 2*. The Tsun = 10 hours [2].

Peak Power Usage (Ppeak):

Maximum prefab units required: $20{,}000 \text{ people} * \frac{1 \text{ prefab unit}}{4 \text{ people}} = 5{,}000 \text{ prefab units}$

How much energy do the prefab units use?

TV: $1 * \left(\frac{50 \text{ W}}{\text{unit}}\right) = 50 \ \frac{\text{W}}{\text{unit}}$

Fans: $2 * \left(\frac{40 \text{ W}}{\text{unit}}\right) = 80 \ \frac{\text{W}}{\text{unit}}$

LED Lights: $5 * \left(\frac{10 \text{ W}}{\text{unit}}\right) = 50 \ \frac{\text{W}}{\text{unit}}$

Refrigerator: $1 * \left(\frac{75 \text{ W}}{\text{unit}}\right) = 75 \ \frac{\text{W}}{\text{unit}}$

$$(50+80+50+75) \frac{\text{W}}{\text{unit}} = 255 \ \frac{\text{W}}{\text{prefab unit}}$$

Energy required for maximum number of prefab units:

$$5{,}000 \text{ prefab units} * \left(\frac{255 \text{ W}}{\text{unit}}\right) * \left(\frac{1 \text{ kW}}{1000 \text{ W}}\right) = 1{,}275.5 \text{ kW}$$

Total Daily Energy Usage (Eused):

Daily hour usage based off assumptions of how long each load type will run as seen on *Table 1*.

TV: $1 * \left(\frac{50 \text{ W}}{\text{unit}}\right) * 6 \text{ hr} = 300 \ W$

Fans: $2 * \left(\frac{40 \text{ W}}{\text{unit}}\right) * 8 \text{ hr} = 1120 \text{ W}$

LED Lights: $5 * \left(\frac{10 \text{ W}}{\text{unit}}\right) * 14 \text{ hr} = 400 \ W$

Refrigerator: $1 * \left(\frac{75 \text{ W}}{\text{unit}}\right) * 24 \text{ hr} = 1800 \ W$

Daily energy usage required for maximum number of prefab units:

$$(300+1120+400+1800)\text{W} = 3620 \text{ W}$$

Cost of Inverters:

$$\text{Cost}_{\text{inverters}} = \text{P}_{\text{peak}} * \frac{\$250}{\text{kW}} = 1286.5\ \text{kW} * \frac{\$250}{\text{kW}} = \$\ 32{,}625$$

Cost of Panels:

$$\text{Cost}_{\text{panel}} = \left(\frac{\text{E}_{\text{used}}}{\text{T}_{\text{sun}}}\right) * \frac{\$350}{\text{kW}} = 18342.5\ \text{kW} * \frac{\$350}{\text{kW}} = \$\ 321{,}625$$

Cost of Batteries:

$$\text{Cost}_{\text{batteries}} = 2 * \text{E}_{\text{used}} * \left(\frac{\$250}{\text{kW}}\right) = 2 * 18342.5\ \text{kW} * \left(\frac{\$250}{\text{kW}}\right) = \$\ 9{,}171{,}250$$

Cost of Batteries, life cycle:

$$\text{Cost}_{\text{batteries,lifecycle}} = \frac{\text{Total}_{\text{system,time}}}{\text{Battery}_{\text{lifetime}}} * \text{Cost}_{\text{batteries}} = \left(\frac{25\ \text{years}}{10\ \text{years}}\right) * \ \$\ 9{,}171{,}250 = \$22{,}928{,}125$$

Cost of Installation:

$$\text{Cost}_{\text{installation}} = \left(\frac{\$0.5}{\text{W}}\right) * \left(\frac{1000\text{W}}{1\ \text{kW}}\right) * \text{P}_{\text{peak}} = \left(\frac{\$0.5}{\text{W}}\right) * \left(\frac{1000\text{W}}{1\ \text{kW}}\right) * 1286.5\ \text{kW} = \$643{,}250$$

Cost Upfront:

$$\text{Cost}_{\text{upfront}} = \text{Cost}_{\text{Inverter}} + \text{Cost}_{\text{Panels}} + \text{Cost}_{\text{Batteries}} + \text{Cost}_{\text{Installation}} = \$10{,}778{,}112$$

Cost of Lifecycle:

$$Cost_{Upfront} = Cost_{Inverter} + Cost_{Panels} + Cost_{batteries,lifecycle} = \$23{,}891{,}737$$

Cost kWh:

$$Cost_{kWh} = \frac{Cost_{Lifecycle}}{9125} * E_{used} = \$\ 0.1427$$

Discussion:

As previously stated, the purpose of is to determine the feasibility of installing a solar PV power plant in a refugee camp in the North of Jordan and compare it to diesel electric generators.

A solar PV plant is made up of several important factors: the PV panels, inverters and the batteries. According to the calculations above, to supply the entire camp with electricity, the PV panels will cost a maximum of $641,987. The inverters will cost a maximum of $321,625 and the batteries will cost $22,928,125. This project costs a maximum of $23,891,737 over the systems lifetime and an upfront cost of $10,778,112.

The solar panels were calculated to have a cost of $0.14/kWh. This value can be compared to diesel generators cost per kWh. On average, a liter of diesel costs about $0.85 in Jordan [3]. The average amount of diesel that diesel generators can produce is 0.4L/ kWh produced [4]. When these two factors are multiplied together, the value turns out to be $0.34/kWh. Since the cost per kWh for the solar panels is significantly less than that of the diesel generators, the PV solar plant would be the more economically favorable choice.

Furthermore, as well as the solar PV power plant being the more economically favorable source of energy, it also holds many advantages when compared to diesel generators. For starters, solar energy is a renewable technology that is very sustainable. Since this project takes place on a refugee camp, many emergencies may take place. For example, vaccine refrigerators need to need to be powered by electricity and cannot always depend on the grid for power in case of an emergency. Solar powered refrigerators have been proven to be a lot more dependable than fuel oils [5]. Also, PV solar power plants are more likely to get financed by climate-related funding because they are more efficient and would reduce the camp's carbon footprint, [6] creating a more appealing argument to the traditional power generation systems. Lastly, it is important to consider that the energy being supplied to a community of people by the generators are very loud, compared to solar panels. It is crucial to carry out this project in the most humane way possible.

Overall, a PV solar power plant is the most economically favorable option because it is cheaper and has many advantages compared to diesel electric operators as previously discussed.

References:

[1] Solargis. “Global Solar Atlas.” *Global Solar Atlas*, globalsolaratlas.info/map?c=32.017392%2C35.809937%2C8&s=32.421703%2C36.095581&m=site.

[2] “Irbid, Jordan - Sunrise, Sunset, and Daylength, November 2019.” *Timeanddate.com*, www.timeanddate.com/sun/jordan/irbid?month=11&year=2019.

[3] “Jordan Diesel Prices, 28-Oct-2019.” *GlobalPetrpPrices.com*, www.globalpetrolprices.com/Jordan/diesel_prices/.

[4] “Diesel Generator.” *Diesel Generator - Energy Education*, energyeducation.ca/encyclopedia/Diesel_generator.

[5] “‘Solarchill-a’ for Medical Purposes.” *Solarchilltests Website*, www.solarchill.org/english/the-solution/preserving-vaccines/.

[6] Leach, Anna. “Clean Energy in Refugee Camps Could Save Millions of Dollars.” *The Guardian*, Guardian News and Media, 17 Nov. 2015, www.theguardian.com/global-development-professionals-network/2015/nov/17/clean-energy-in-refugee-camps-could-save-millions-of-pounds.